Special Topics in Technology

Introduction to Nanotechnology

Jason Ling-Wei Luo

Department of Physics, National Tsing Hua University

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Outline

1 Self Introduction

2 Introduction
- Length Scale
- Nanoscience and Nanotechnology
- Some general Tools
- Techniques
- How is the Nanostructures

3 Some Special Effects
- Lotus Effect
- Small Size Effect

4 Outlook

5 Reference

Outline

1 Self Introduction

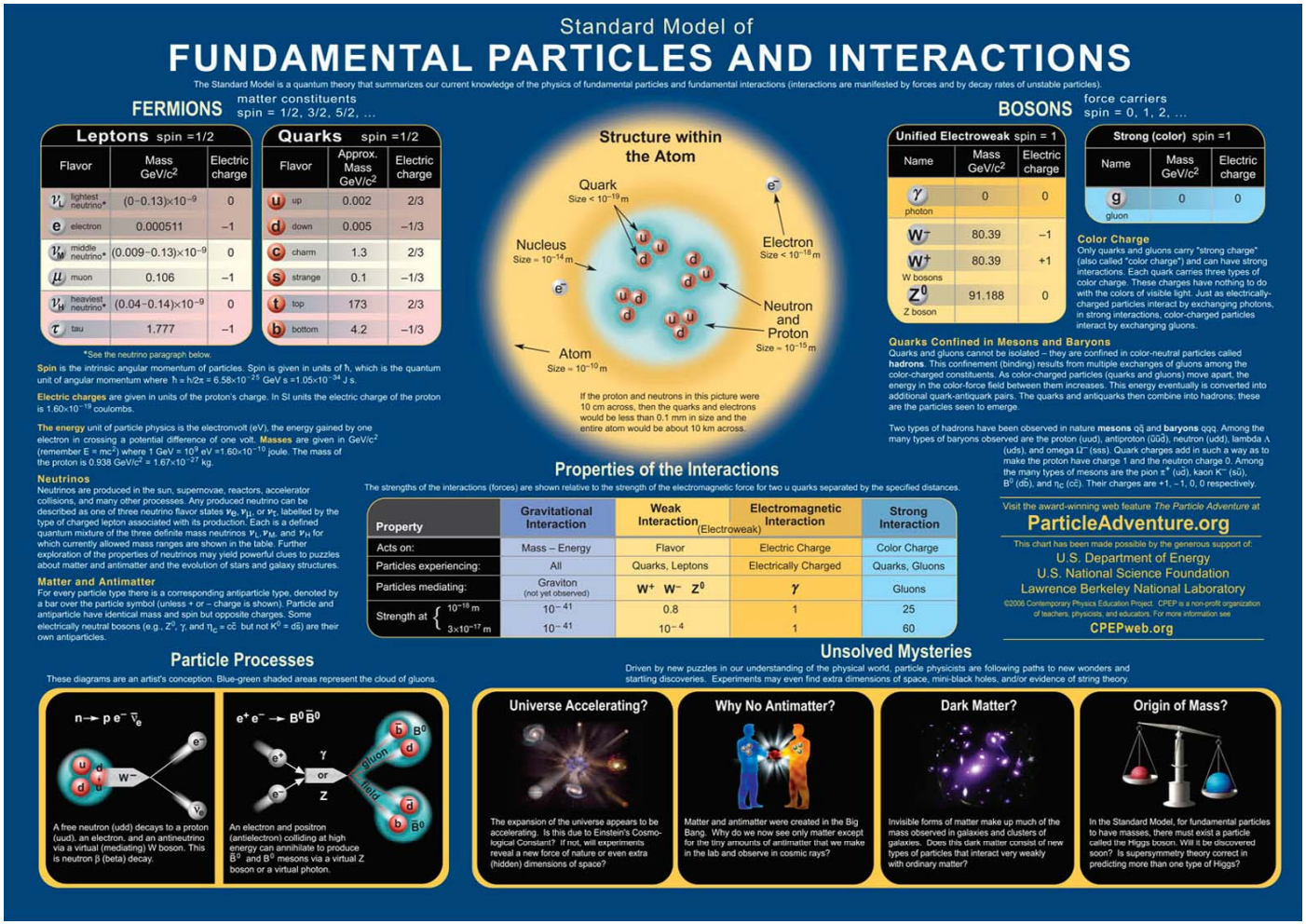
2 Introduction

3 Some Special Effects

4 Outlook

5 Reference

- Jason Ling-Wei Luo
- 2nd year Ph. D. student in Department of Physics
- Particle physics and Cosmology



Outline

1 Self Introduction

2 Introduction

3 Some Special Effects

4 Outlook

5 Reference

Outline

1 Self Introduction

2 Introduction
- Length Scale
- Nanoscience and Nanotechnology
- Some general Tools
- Techniques
- How is the Nanostructures

3 Some Special Effects

4 Outlook

5 Reference

Length Scale

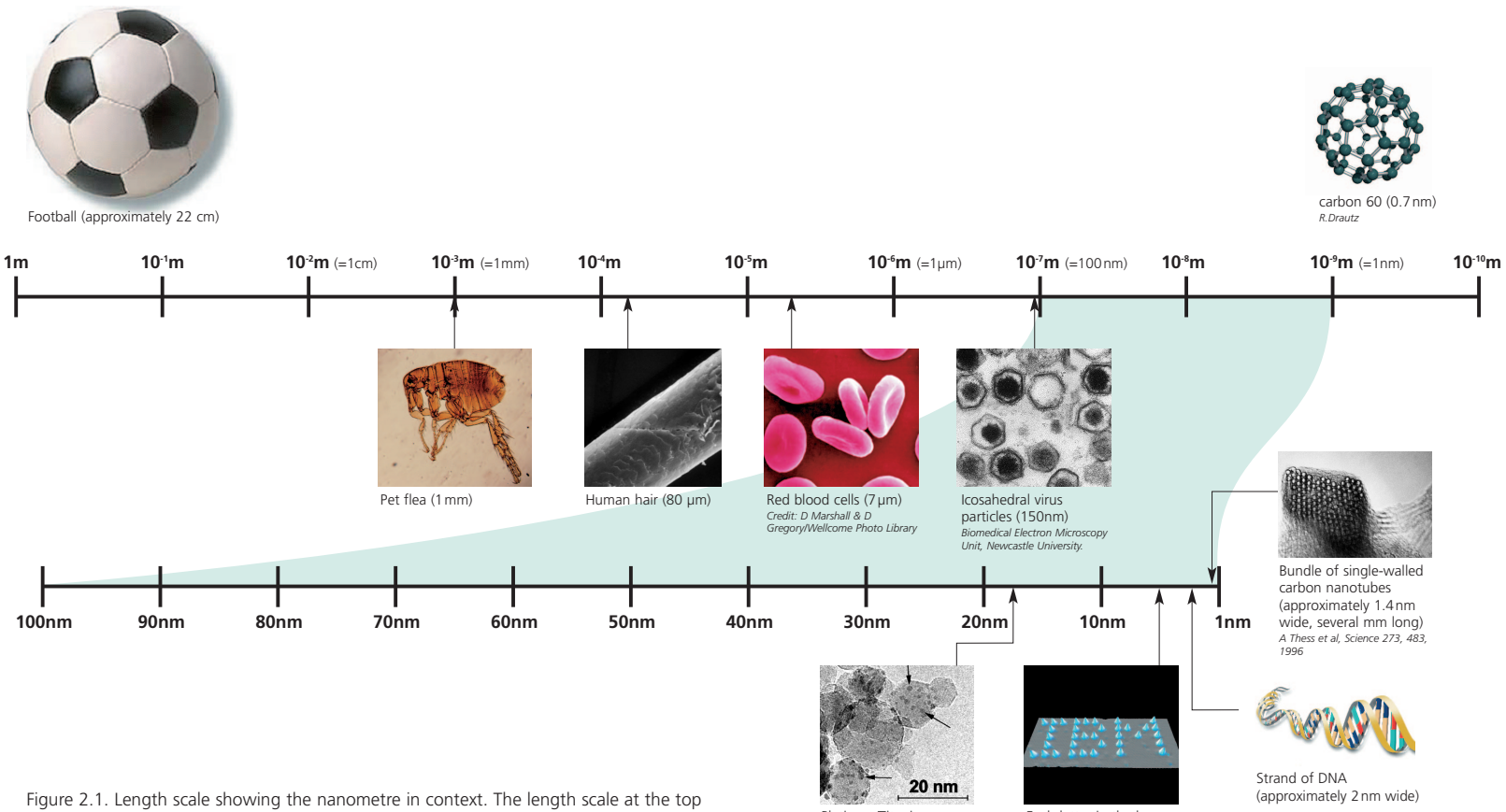


Figure 2.1. Length scale showing the nanometre in context. The length scale at the top ranges from 1m to 10^{-10}m, and illustrates the size of a football compared to a carbon 60 (C_{60}) molecule, also known as a buckyball. For comparison the world is approximately one hundred million times larger than a football, which is in turn one hundred million times larger than a buckyball. The section from 10^{-7}m (100 nm) to 10^{-9}m (1 nm) is expanded below. The lengthscale of interest for nanoscience and nanotechnologies is from 100 nm down to the atomic scale - approximately 0.2 nm.

Classification of Scale

- Macroscopic scale: > 100nm
- Microscopic scale: < 1nm
- Mesoscopic scale: 1 to 100nm

Outline

1 Self Introduction

2 Introduction
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- Nanoscience and Nanotechnology
- Some general Tools
- Techniques
- How is the Nanostructures

3 Some Special Effects

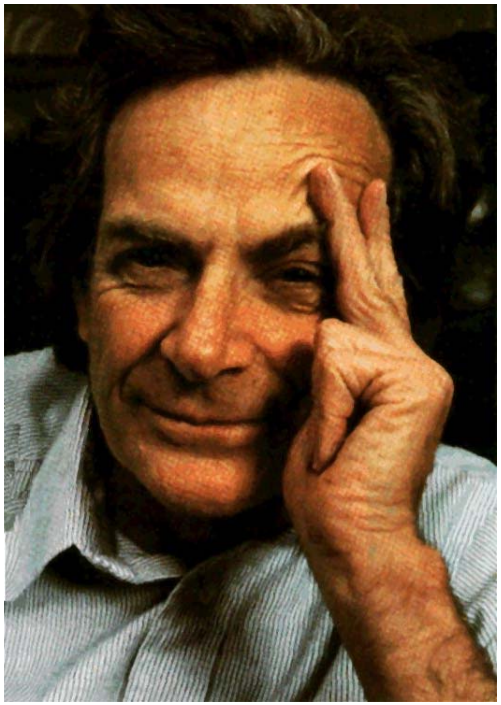
4 Outlook

5 Reference

Nanoscience and Nanotechnology

- Nanotechnology develops quickly in recent years and is getting more and more popular.
- What is a mysterious field in the size of nanometer?
- In that size, many things can not understand by our eyes, you can see something in the size with the microscope only.
- It is the technique about the atomic and molecular scale.
- Nanotechnology can be used in many fields and covers from physics, chemistry, materials science, electro-optical science, biology, and medicine etc.

The Father of Nanotechnology



- In 1959, it is the fist time to use the word 'nano-technology' in a talk "There's Plenty of Room at the Bottom" given by Prof. Richard Feynman at Caltech.
- Prof. Feynman has predicted that people can would develop the technique to manipulate individual atoms and molecules.

Outline

1 Self Introduction

2 Introduction
- Length Scale
- Nanoscience and Nanotechnology
- Some general Tools
- Techniques
- How is the Nanostructures

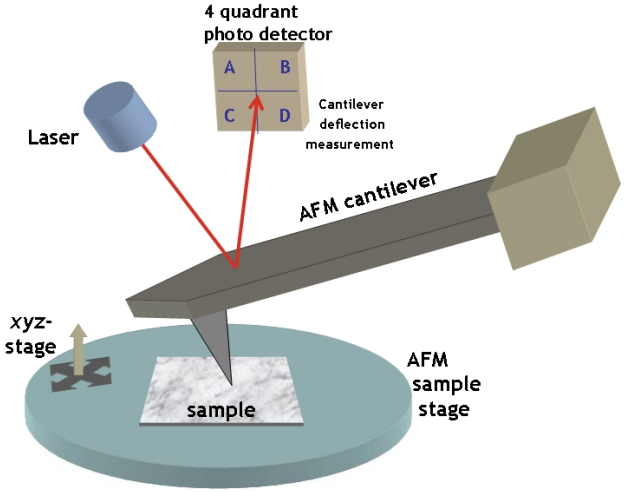
3 Some Special Effects

4 Outlook

5 Reference

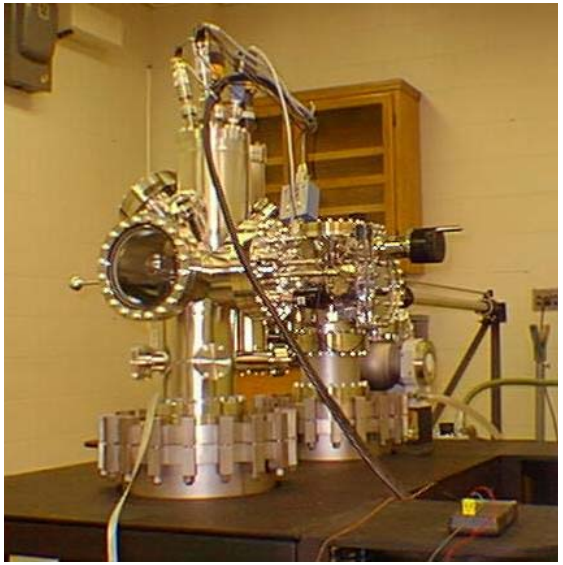
Some general Tools

- Microscopy would provide people a method to understand the nano scale world.
- Microscope
  - Non-optical microscope: use a tiny tip to scan the surface profile of the materials, e.g. SPM.
    - AFM: due to Van der Waals force
    - STM: due to quantum tunneling effect

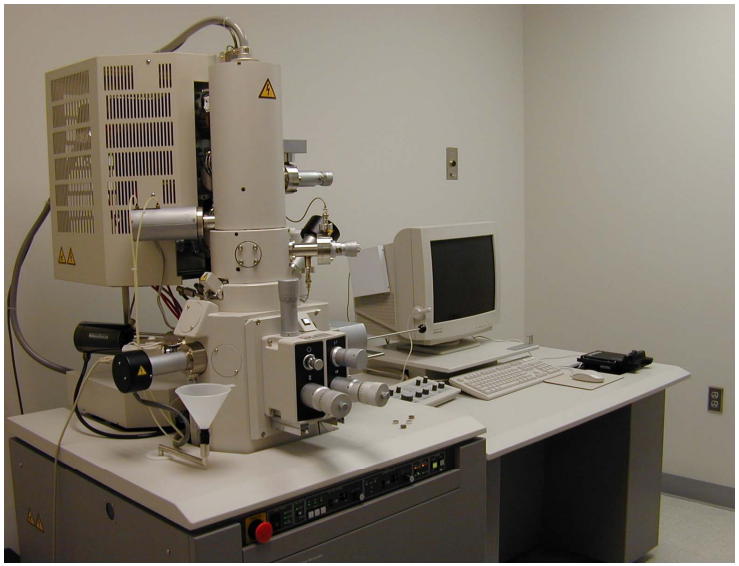


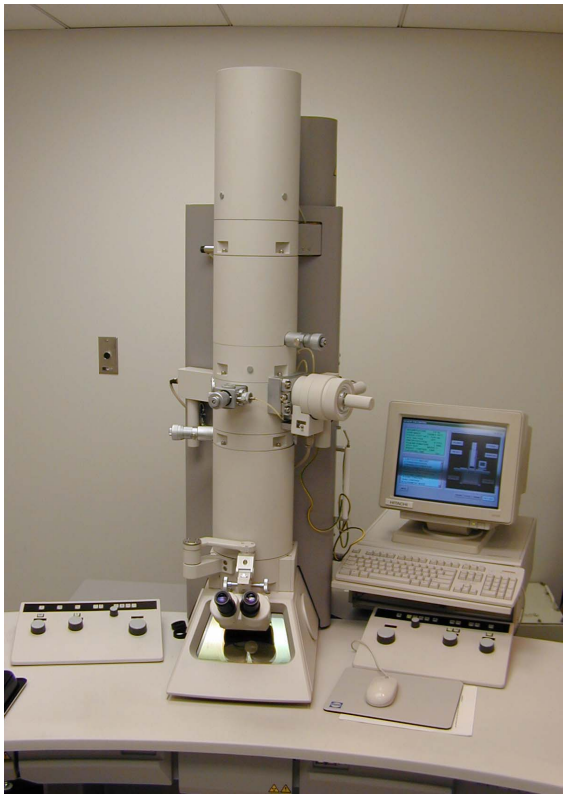
  - Optical microscope: application of "wave–particle duality", e.g. SEM and TEM.

AFM and STM



SEM and TEM





Outline

1 Self Introduction

2 Introduction
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- Nanoscience and Nanotechnology
- Some general Tools
- Techniques
- How is the Nanostructures

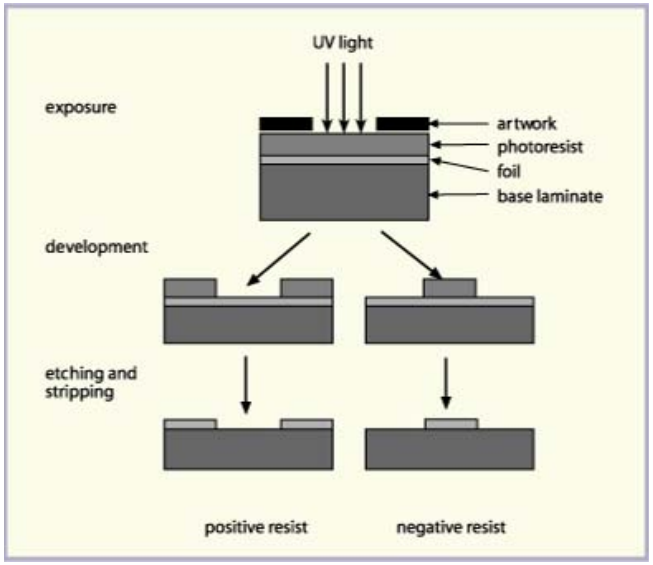
3 Some Special Effects

4 Outlook

5 Reference

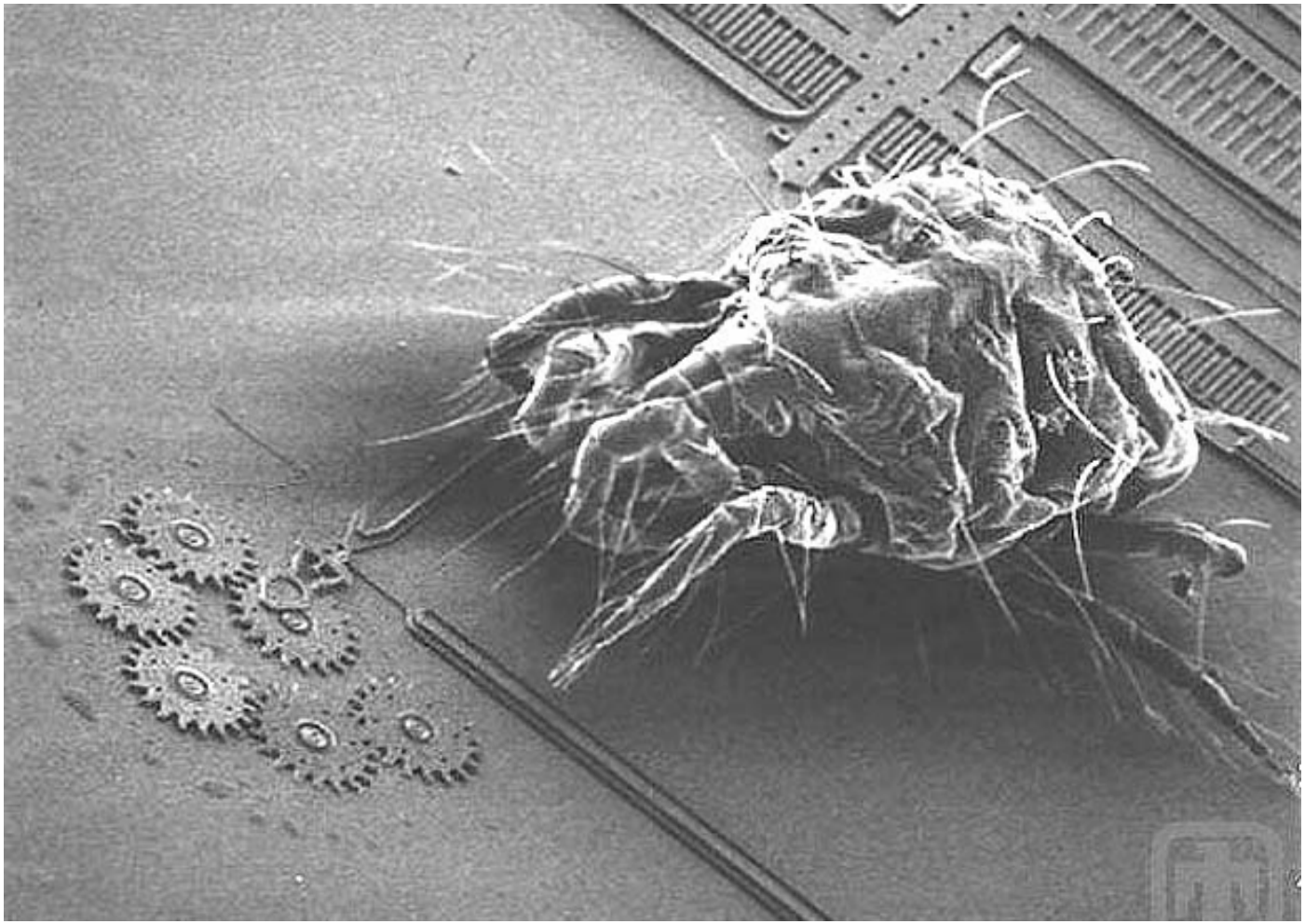
Techniques

- Micro-electro-mechanical system (MEMS) are made up of components between 1 to 100μm in size and MEMS devices generally range in size from 20μm to 1mm.



- Photolithography (sometimes just called lithography)
  - Transfer of a pattern into a photosensitive material by selective exposure to a radiation source such as light.
  - It is used in semiconductor manufacturing process extensively.

Mite and MEMS



Outline

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- Some general Tools
- Techniques
- How is the Nanostructures

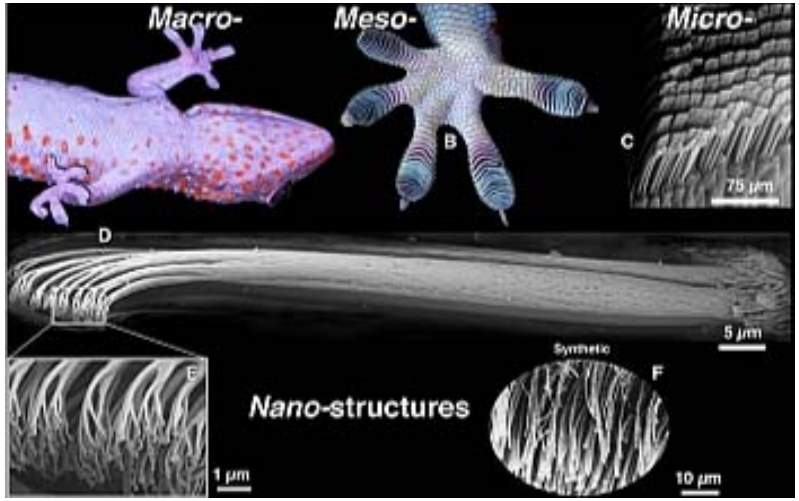
3 Some Special Effects

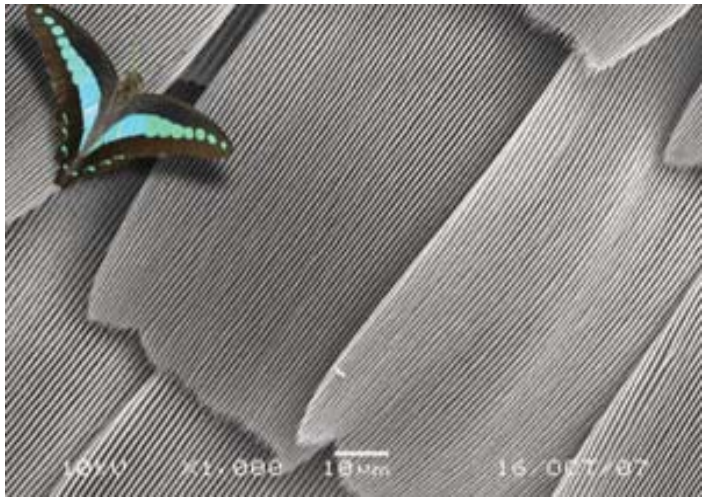
4 Outlook

5 Reference

Some Nanostructures

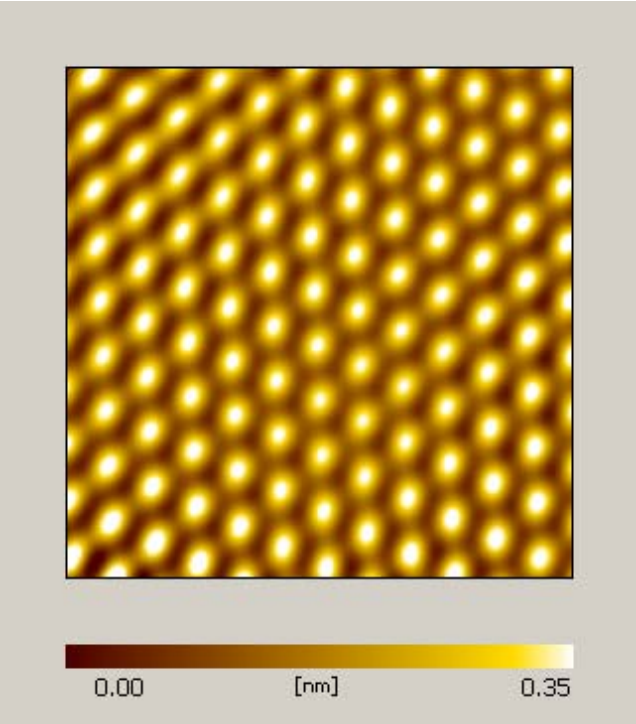
- The gecko foot: the structure of the foot relies on the millions of sub-microscopic branched hairs called setae.
- The beautiful butterfly: the nanostructure would change the optical properties of the surface

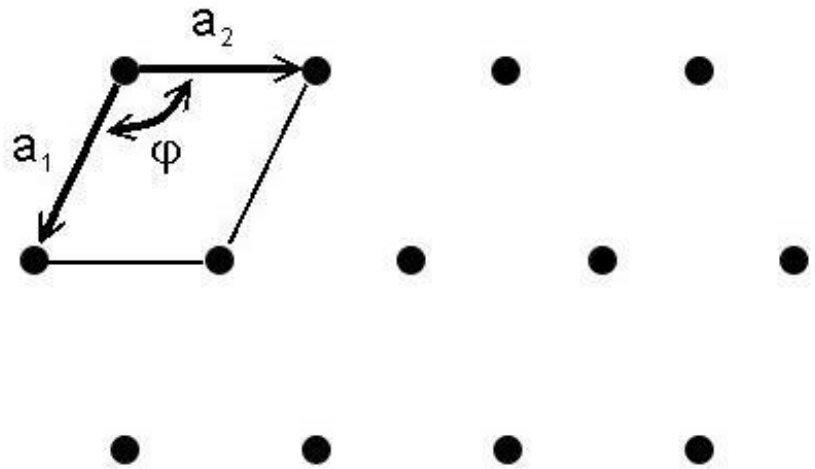




AFM Surface Profile

- Atom of mica (Hexagonal Close-Packed Structure)

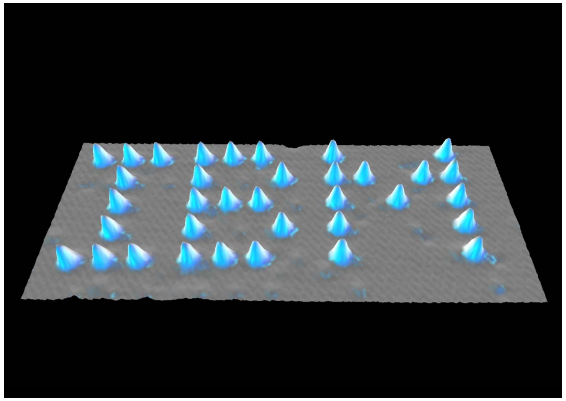


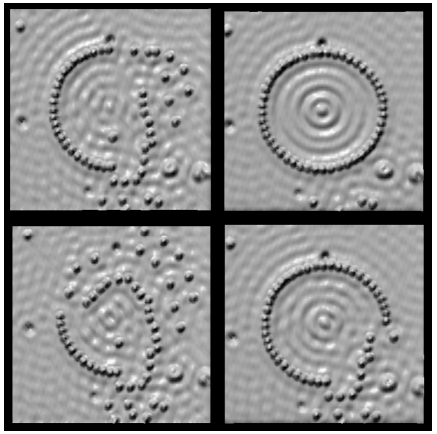


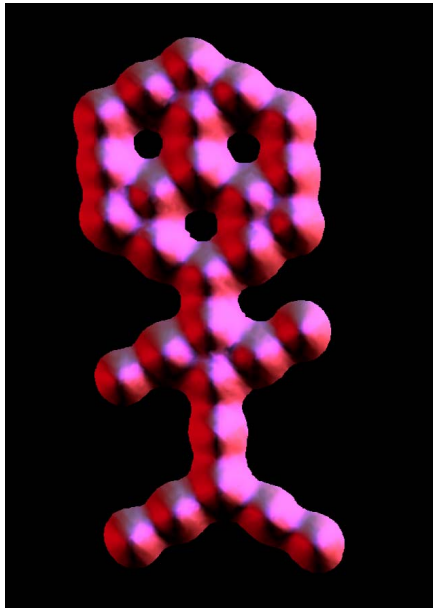
Hexagonal lattice

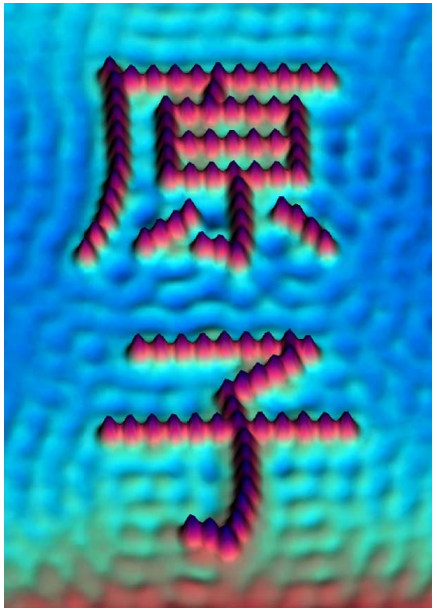
$|a_1| = |a_2|$; $\varphi = 120^0$

STM Surface Profile

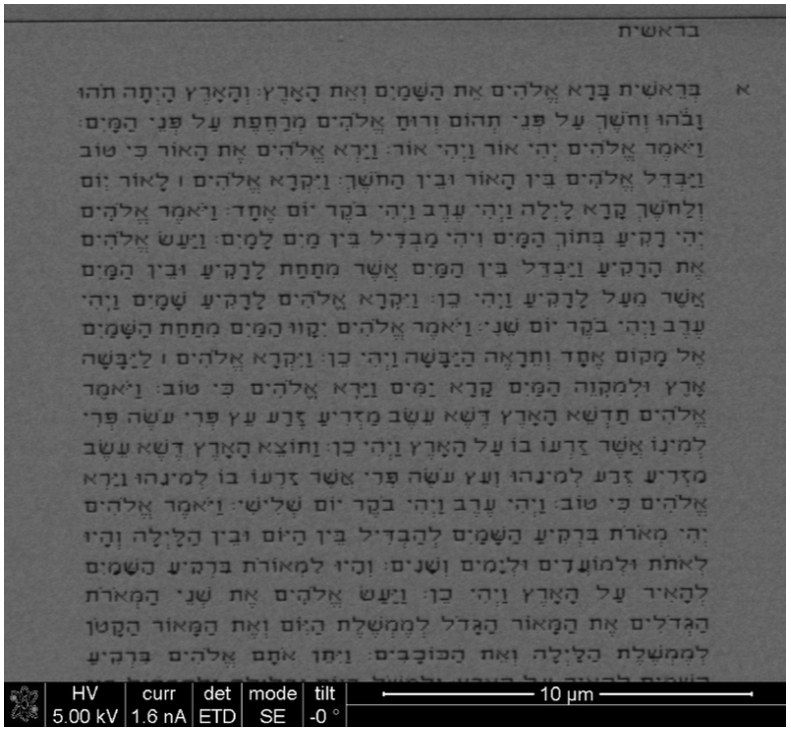








Nano-Bible



Outline

1 Self Introduction

2 Introduction

3 Some Special Effects

4 Outlook

5 Reference

Outline

1 Self Introduction

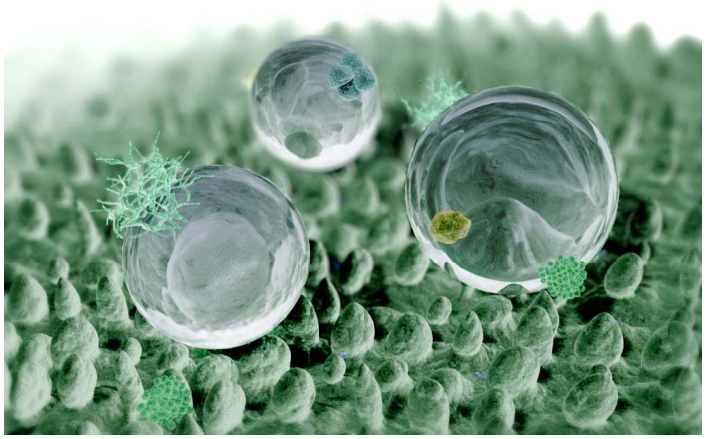
2 Introduction

3 Some Special Effects
- Lotus Effect
- Small Size Effect

4 Outlook

5 Reference

Lotus Effect



- It is the the famous effect of nanotechnology.
- It is said that the materials just cover with some nanostructures and would not be dirty, a self-cleaning properties.

Outline

1 Self Introduction

2 Introduction

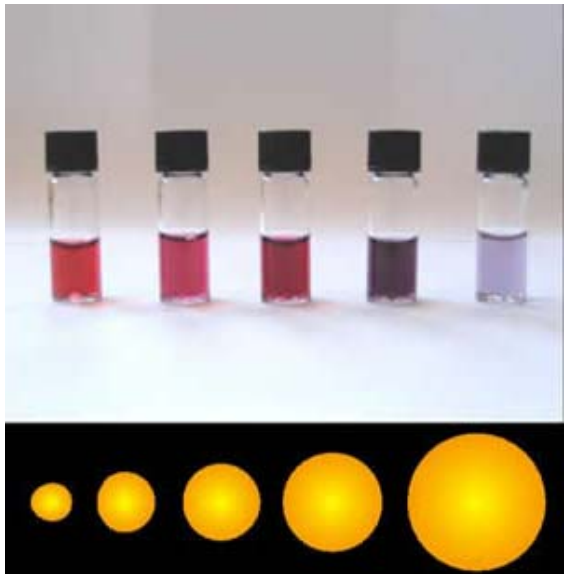
3 Some Special Effects
- Lotus Effect
- Small Size Effect

4 Outlook

5 Reference

Small Size Effect

- It means that the size of the material getting small and small would change some of its physical properties.



- The diameter of gold nanoparticles determines the wavelengths of light absorbed.

Outline

1 Self Introduction

2 Introduction

3 Some Special Effects

4 Outlook

5 Reference

Outlook

- The progress of technology is an important thing that represents the world is advance or stagnant.
- Anything is dependent on technology to get improved.
- People are devoting to understand the properties of the nanostructures to improve our life.
- Nanotechnology is becoming a part of our life.

End

- Thanks for your attention!

Outline

1 Self Introduction

2 Introduction

3 Some Special Effects

4 Outlook

5 Reference

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- μmasch website, http://www.spmtips.com/.
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- Autumn Lab website, http://lclark.edu/∼autumn.